

Minor Project

1010063491

A.L.E.R.T.

Project ID	CE_AIML_31
Enrollment No	2301030600043, 2301030600045, 2301030600049
Name	Prem Deshani, Kavya Kothari, Mehulsinh Rathod
Branch	BTech. CE-AIML
Guide Name	Ms Vishwa Patel

Index

- Introduction
- Background and Motivation
- Relevance and Importance
- Literature review
- Objectives
- Required Tools and Technology
- Method/ Approach
- Bibliography

Introduction

- **Automotive Safety Crisis:** Driver fatigue causes over 20% of commercial vehicle crashes worldwide, making real-time Driver Monitoring Systems (DMS) essential.
- **Facial Behavioral Analysis:** Uses non-invasive computer vision to analyze eye openness (EAR), yawning (MAR), PERCLOS, and head pose angles.
- **The Non-IoT Edge Approach:** Executes 100% on-device AI inference directly on embedded hardware (Raspberry Pi 5 / NVIDIA Jetson Orin Nano).
- **Key Technical Advantages:** Delivers sub-5ms latency, complete offline reliability, zero cellular bandwidth cost (\$0/mo), and total data privacy (GDPR compliance).

Background and Motivation

- **Flaws of Cloud/IoT Architectures:** Cloud-dependent systems suffer from network latency jitter (150ms–500ms+), rendering micro-sleep alerts ineffective.
- **Connectivity Dead Zones:** Vehicles frequently navigate tunnels, mountain passes, and rural highways where 4G/5G cellular signal is lost.
- **Biometric Privacy Liabilities:** Transmitting high-definition facial video streams to cloud servers poses severe GDPR and biometric privacy risks.
- **High Operating Expense:** Cloud video streaming costs \$15–\$30/month per vehicle in cellular data and cloud server compute fees.
- **Edge Motivation:** An autonomous, sub-5ms on-device system guarantees 100% alert reliability anywhere without recurring costs.

Relevance and Importance

- **Sub-5ms Compression Pipeline:** Combines MediaPipe BlazeFace + Face Mesh backbones with Quantization-Aware Training (QAT INT8) for sub-5ms inference.
- **24/7 NIR Lighting Robustness:** Implements 850nm / 940nm Near-Infrared illumination with adaptive CLAHE contrast enhancement for day & night operation.
- **Eyewear & Sunglasses Robustness:** Features dual-wavelength NIR penetration and dynamic fallback to Mouth Aspect Ratio (MAR) + Head Nodding when eyes are occluded.
- **Personalized Calibration (<10s):** Automatically calculates baseline thresholds ($\text{EAR_thresh} = \text{mean} - 2.5 * \text{std}$) without manual configuration.
- **Commercial Relevance:** Reduces fleet accident rates, lowers insurance liabilities, and scales easily across commercial trucking, buses, and ride-shares.

Literature Review

- **Soukupova & Cech (2016) [R1]:** Introduced Eye Aspect Ratio (EAR) using 68 facial landmarks for real-time blink and eye closure detection.
- **Dinges & Grace (1998) [R2]:** Established PERCLOS (Percentage of Eye Closure $\geq 80\%$ over time window) as the gold standard fatigue metric.
- **Garces et al. (2021) [R5]:** Formulated Mouth Aspect Ratio (MAR) to quantify yawning magnitude and duration.
- **Dzyakanchuk et al. (2022) [R3]:** Evaluated low-power CNN face detection on Raspberry Pi embedded hardware.
- **Research Gap Addressed:** Existing literature lacks sub-5ms INT8 quantized multi-metric landmark pipelines with NIR robustness on sub-\$100 edge hardware.

Objectives

- To prevent fatal road accidents and save lives by detecting driver fatigue and micro-sleeps in real time before loss of vehicle control occurs.
- To develop a completely non-intrusive safety system that monitors driver alertness using standard camera vision without requiring wearable sensors or manual driver setup.
- To ensure 100 percent offline and privacy-preserving operation, enabling reliable fatigue monitoring on highways and tunnels without relying on cellular networks or cloud infrastructure.
- To build an affordable, edge-deployable solution that can be easily integrated into commercial trucking fleets, public transport, and consumer vehicles.

Required Tools and Technology

- **Hardware Platforms:** Raspberry Pi 5 , Google Coral USB Edge TPU (\$60), Hailo-8L 13 TOPS HAT, NVIDIA Jetson Orin Nano.
- **Optical Hardware:** Near-Infrared (NIR) 850nm/940nm LED arrays, narrow bandpass optical filters, No-IR-Cut camera sensor.
- **Programming Runtimes:** Python 3.10+ & C++17 for high-performance multithreaded frame ingestion and inference.
- **Landmark Extraction:** MediaPipe FaceLandmarker Tasks API — 468 3D landmarks, 6.7MB TFLite model, ~8–15ms inference via XNNPACK CPU delegate, LIVE_STREAM async mode at 640×480.

Required Tools and Technology

- **Computer Vision:** OpenCV 4.8+ — camera I/O (MJPG codec, 30 FPS), solvePnP head pose, eye/mouth ROI cropping, HUD rendering.
- **Classical ML (Model A):** LightGBM 4.0+ — gradient boosted trees on 9D feature vector; scikit-learn for preprocessing pipelines and Random Forest POC baseline.
- **Deep Learning (Models B & C):** PyTorch 2.1+ — GRU sequence model (Model B) and CNN eye-patch classifier (Model C); ONNX Runtime 1.16+ for cross-platform edge serving.
- **Feature Engineering:** NumPy (rolling window arrays, EAR/MAR geometry), SciPy (EMA drift adaptation), imbalanced-learn (SMOTE for L3 class imbalance).
- **Data Processing:** Pandas (dataset loading, label alignment across UTA-RLDD / NTHU-DDD / YawDD / MRL Eye), OpenCV VideoCapture (frame extraction from video datasets).
- **Dev Environment:** Python 3.11 in virtualenv, VS Code, Git — all inference ONNX-exportable and forward-compatible with OpenVINO (Intel NPU) and TensorRT INT8 (Jetson).

Required Tools and Technology

- **UTA-RLDD:** 60 subjects, ~30 min per subject, ~6.4M total frames, 95 GB. Naturalistic webcam recordings in home/office environments. Labels: Alert, Low Vigilance, Drowsy (maps to L0, L1, L2). Primary training set.
- **NTHU-DDD:** 36 subjects, ~14 GB. Recorded under 4 conditions: day/night crossed with glasses/no glasses. Frame-level binary labels: Drowsy or Non-Drowsy. Added specifically for glasses occlusion and low-light robustness.
- **YawDD:** 107 subjects, ~8 GB. Dual-camera in-vehicle recording from an actual dashcam mounted below eye level inside a car. Labels: Normal, Talking, Yawning. Added to train the model on the real dashcam perspective that desk-webcam datasets cannot provide.
- **MRL Eye Dataset:** 37 subjects, 84,898 labelled eye images, ~2 GB. Covers 8 image scales, varied lighting including infrared, with and without glasses. Binary labels: open or closed per image. Used exclusively to train Model C, the CNN eye-patch classifier.

Method/ Approach

- **Stage 1** - Face and Landmark Extraction: Each 640x480 frame is passed to MediaPipe FaceLandmarker, yielding 468 3D landmark coordinates. 24 anatomically selected points are extracted: 6 per eye for EAR, 8 mouth contour for MAR, and 6 solvePnP anchors (nose tip, chin, eye corners, mouth corners).
- **Stage 2** - Geometric Feature Computation: Instantaneous EAR, MAR, and MOE are computed per frame. solvePnP maps the 6 anchor landmarks to a canonical 3D face model to extract Yaw, Pitch, and Roll. EAR is corrected for head-turn foreshortening using the yaw angle.
- **Stage 3** - Personalised Calibration (Session Start, 10s): 300 frames of the alert driver are collected at session start. Per-driver EAR and MAR thresholds are computed from mean and standard deviation, eliminating false alarms from natural anatomical variation such as narrow eye shapes.

Method/ Approach

- **Stage 4** - Temporal Feature Engineering: A 9-dimensional feature vector (EAR, EAR std dev, PERCLOS, Blink Rate, Blink Duration, MAR, Yawn Count, Head Pitch, MOE) is assembled using 5-second short-term and 60-second long-term rolling windows. A slow exponential moving average continuously adapts the baseline for lighting drift over long drives.
- **Stage 5** - Ensemble Classification: Model A (LightGBM) classifies the 9D vector per frame with a weight of 0.50. Model B (GRU, 32 hidden units) classifies a 5-second sequence of 4 key features every 5 seconds, detecting gradual fatigue trends invisible to per-frame classifiers, with a weight of 0.30. Model C (tiny CNN) classifies a raw 32x16 pixel eye crop per frame, independent of landmark accuracy, providing redundancy under glasses or glare, with a weight of 0.20. The three probability outputs are fused via weighted soft vote to produce a final L0 to L3 prediction. A hard safety override forces L3 if severe eye closure is detected regardless of the fusion result.
- **Stage 6** - Alert Output and HUD: Fatigue level drives tiered alerts (L1 amber chime, L2 repeated alarm, L3 critical alert). A 30 FPS overlay displays live fatigue level, per-model probabilities, all 9 signal values, and calibration status on the camera stream.

Bibliography

- [1] Soukupova, T., & Cech, J. (2016). Real-Time Eye Blink Detection Using Facial Landmarks. Computer Vision Winter Workshop.
- [2] Dinges, D. F., & Grace, R. (1998). PERCLOS: A Valid Psychophysiological Measure of Alertness. US Dept of Transportation.
- [3] Dzyakanchuk, A., et al. (2022). Low-Power Face Detection on Raspberry Pi for Automotive DMS. IEEE Access.
- [4] Acharjee, S., et al. (2020). Edge-Based Driver Drowsiness Monitoring on Embedded Hardware. SN Computer Science.
- [5] Garces, M., et al. (2021). Automated Yawn Detection Using Mouth Aspect Ratio. IEEE Sensors Journal.
- [6] MediaPipe Face Mesh Topology (468 3D Landmarks). Google AI Edge Documentation (2023).
- [7] NVIDIA TensorRT INT8 Developer Guide for Jetson Orin Platform (2024).
- [8] Hailo-8L Dataflow Compiler & Software Suite User Guide (2024).